Multilayer Global Spectral–Spatial Attention Network for Wetland Hyperspectral Image Classification

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***Abstract*— Coastal wetland monitoring plays an important role in the protection and restoration of ecosystems in this world. UAV-hyperspectral imaging, as an emerging technique for Earth observation and space exploration, provides the huge potential ability to identify different wetland species. In this work, a multilayer global spectral–spatial attention network (MGSSAN) is proposed for mapping coastal wetlands, which mainly consists of two major steps. First, a two-branch convolutional neural network (CNN) framework with residual connection is developed to obtain an initial classification probability map, in which one branch is used to capture the spectral information, the other branch is used to extract spatial information, and a global spectral–spatial attention module is designed to guide networks focusing on those features that are more discriminative. Second, an extended random walker method is utilized to optimize the initial classification probabilities, so as to yield the final map. Experiments performed on three wetland HSI datasets created by ourselves verify that the proposed method can obtain superior performance with respect to several state-of-the-art hyperspectral image classification methods.**

***Index Terms*— Deep learning, extended random walk, global spectral–spatial attention (GSSA), hyperspectral image classification, residual connection.**

I. Introduction

WETLANDS are the most important ecosystem in maintaining biodiversity and ecological balance. Classification of coastal wetlands can provide essential reference information for formulating wetland protection and protecting wetland species diversity. Hyperspectral images (HSIs) can not only capture the detailed spectral information but also the spatial structures of different land covers and, thus, quite useful in distinguishing different wetland species [1]–[6].

In recent years, with the development of remote sensing observation technology, the amount of hyperspectral data acquired by different platforms, such as satellites, airplanes, and unmanned aerial vehicle (UAV) observation platforms, is gradually increasing. Compared with other observation platforms, the UAV platform has the characteristics of high flexibility, strong maneuverability, and simple operation [7], [8]. Due to these advantages, UAV hyperspectral remote sensing has become an easy and fast Earth observation technology.

A lot of studies have been carried based on the hyperspectral imaging system on UAV drones. Zhong *et al.* [9] constructed a hyperspectral agricultural remote sensing dataset based on the UAV platform, which is used for precise crop classification. Li *et al.* [10] established a classification framework for rapid survey and update of wetland environment using UAV system, which provides a solid foundation for wetland biological invasion monitoring. Ma *et al.* [11] analyzed the distribution and change trend of coastal wetlands through UAV hyperspectral remote sensing monitoring technology and provided essential support for the protection and management of coastal wetlands.

In the preliminary stage, most of the HSI classification methods are based on spectral features, such as support vector machine (SVM) [12], random forest (RF) [13], and multinomial logistic regression (MLR) [14]. Nevertheless, these approaches easily result in noisy labels on the classification map because of ignoring spatial information. To overcome this issue, many researchers pay more attention to spectral–spatial classification. For example, Benediktsson *et al.* [15] constructed morphological features (MPs) with a series of opening and closing operations and, then, used MP for HSI classification. Following this work, a variety of MP methods have been investigated, such as extended morphological attribute profiles (EMAPs) [16], parameter-adaptive EMAPs [17], and random subspace ensemble-based EMAPs [18]. Li and Du [19] used the Gabor filter to extract spatial features from HSI to improve classification accuracy. Kang *et al.* [20] successfully

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applied edge-preserving filtering technology to extract spatial features of HSIs for the first time. Duan *et al.* [21] constructed structural profiles to extract discriminative features under complicated imaging scenes. These methods mentioned above extract spectral–spatial features with the guidance of prior information and manual setting of empirical parameters and, thus, are not able to obtain optimal classification performance on different scenes [22].

Recently, deep learning methods have shown their potential in many fields, such as object detection, image classification, and natural language processing. In the HSI classification community, convolutional neural networks (CNNs) [23]–[27], recurrent neural networks [28]–[31], and generative adversarial networks [32]–[34] have been widely used for ground object identification. For example, Chen *et al.* [23] introduced CNN as a feature extractor of hyperspectral classification. Yu *et al.* [35] proposed an end-to-end CNN for hyperspectral classification (2DCNN). Due to the abundant spectral information, it is important to extract comprehensive spectral features. Therefore, Hamida *et al.* [36] proposed a 3DCNN to enable the combination of spectral and spatial information. Compared with 2DCNN, 3DCNN can obtain higher classification performance while bringing higher computational complexity. Roy *et al.* [37] proposed a 3-D–2-D CNN for spectral–spatial feature extraction, which can combine the advantages of the two types of network and improve the classification performance without increasing model complexity. The methods mentioned above are based on single-branch networks and, thus, cannot characterize different spectral and spatial features of HSIs simultaneously.

In order to address this problem, some researchers focus on extracting spatial and spectral features separately through two-branch networks. For instance, Yang *et al.* [38] extracted the spatial and spectral features via a two-branch network, and then, the spatial and spectral features are concatenated to fully connected layers for classification. Kang *et al.* [39] proposed a novel dual-path network, which combines the advantages of the residual network and the dense convolutional network. Xu *et al.* [40] designed a two-branch network to capture spatial–spectral features, in which a long short-term memory (LSTM) structure is used to extract spectral information, and a CNN is used to extract spatial information. Li *et al.* [41] utilized different receptive fields to extract global and local information from HSIs.

These methods mentioned above use all spectral–spatial features for classification without considering the high redundancy of these features. To reduce the useless information and preserve important features, some researchers introduced attention mechanisms [42], [43] in the process of feature extraction. For example, Ma *et al.* [44] proposed a double-branch multiattention mechanism network (DBMA) for the classification of HSIs. Wang *et al.* [45] proposed a band-attention network that can select significant bands and discard useless bands for hyperspectral classification. Inspired by the DBMA algorithm, Li *et al.* [46] proposed a double-branch dual-attention mechanism network that includes an adaptive self-attention mechanism. Guo *et al.* [47] directly applied the attention mechanism to the original HSI input and adopts a feature grouping processing strategy to make full use of the redundancy and differences of different channels so that they can acquire finer features. Hang *et al.* [48] developed a novel spectral and spatial attention mechanism to the two branch networks. Many studies have demonstrated that deep neural networks with attention mechanisms are beneficial for land cover classification.

In this work, to efficiently utilize the spectral and spatial information at different layers, a multilayer global spectral–spatial attention network (MGSSAN) is proposed for mapping coastal wetlands, which mainly consists of three key steps. First, a two-branch CNNs with the global spectral–spatial attention (GSSA) module is proposed to obtain an initial class probability map. Then, the extended random walker (ERW) is exploited to optimize the initial probability map by considering the correlation among neighboring pixels. Experiments on the wetland hyperspectral datasets demonstrate that the proposed method outperforms other state-of-the-art classification approaches. The major contributions of this work can be concluded as follows.

1) To meet the requirement of wetland monitoring, a new hyperspectral database in the coastal wetland is constructed, named the YRD dataset, which is a UAV-borne hyperspectral dataset with high spatial and spectral resolutions, collected in the Yellow River Delta National Nature Reserve in Shandong Province, China.
2) We design a novel GSSA mechanism to learn the discriminative features from shallow to deep layers, which can well preserve useful information and suppress useless ones.
3) We propose an attention-guided two-branch CNNs' classification framework. Different from other spectral–spatial classification methods, we incorporate the GSSA module to each convolutional layer, so as to fully exploit the spectral–spatial information at different layers.

The rest of this article is organized as follows. Section II describes the study regions and the YRD dataset. Section III presents the proposed classification framework in detail. Experimental results and analysis of MGSSAN are reported in Section IV. Finally, the conclusions and discussions are summarized in Section V.

II. Materials

A. Study Regions

The Yellow River coastal wetland is a rare estuary wetland ecosystem in the world, located on the coast of the Bohai Sea in the northeast of Shandong Province, China (36°55′–38°16′N,117°31′–119°18′E), as shown in Fig. 1. Vegetation coverage is as high as 53.7%, which is the largest beach nature vegetation area in the coastal area of China. In addition, there are also about 4.5 million birds in the national natural reserve. Due to the low terrain, seawater will flood coastal areas during high tide, which causes soil salinization and provides a good living environment for Spartina alterniflora. This also leads to the fact that the Spartina alterniflora is increasingly threatening the ecological pattern of native vegetation (Reed, Tamarix, and Suaeda glauca) and destroys the ecological balance and species diversity.

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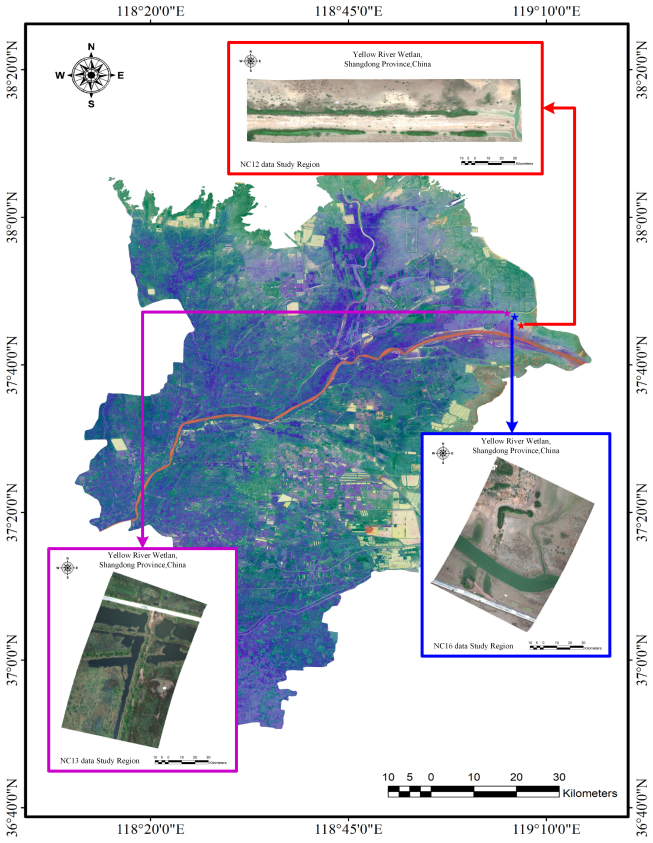


Fig. 1. Study regions.

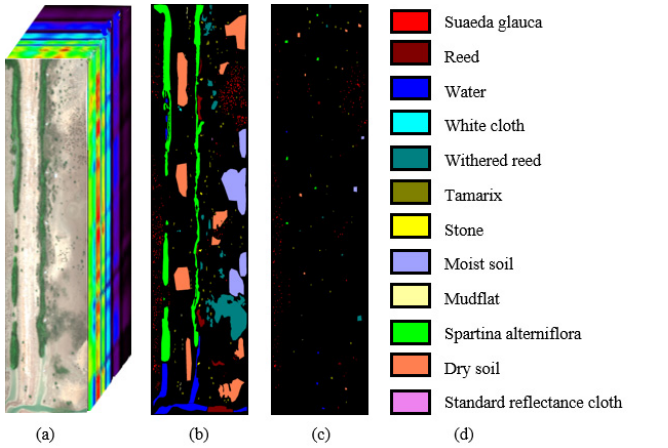


Fig. 2. NC12 dataset. (a) Image cube. (b) Ground-truth image. (c) Training data visualization. (d) Class name.

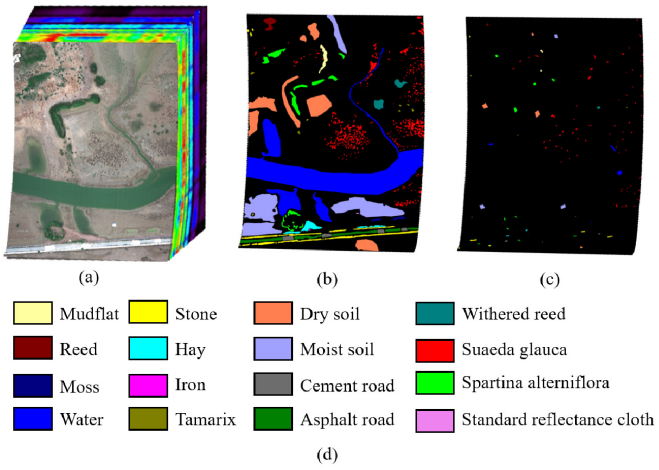


Fig. 3. NC16 dataset. (a) Image cube. (b) Ground-truth image. (c) Training data visualization. (d) Class name.

B. Data Preprocessing

In this study, we constructed a new database for wetland land-covers classification named the Yellow River Delta (YRD) HSI datasets.[1] All datasets were acquired by a 12-mm focal length Headwall Nano-hyperspec imaging sensor equipped on DJI Matrice 600 (DJI M600) UAV platform, and the sensor records 270 bands from 400 to 1000 nm. We process the UAV hyperspectral data with the Hyperspec software provided by the instrument manufacturer, and the main preprocessing steps include radiometric calibration, reflectance correction, and geometric correction. For radiometric calibration, the original DN value collected by the HSIr is converted into radiance data, by loading the radiance calibration configuration file corresponding to the integration time when leaving the scene, which can be written as

$$\mathrm{DN} = L_1 * G * \mathrm{t_{exp}} + \mathrm{DF} \tag{1}$$

where DN is the original light intensity recorded by the hyperspectral imaging sensor, L_1 is the radiance value (only the light splitting of the first grating is considered), G is the camera gain, t_{exp} is the integration time of the photosensitive element, and DF is the dark current noise. A standard reflectance cloth is used as a reference for reflectance correction, which contains 56%, 32%, and 11% three standard reflectances. Then, we select an unexposed area on the reflectance cloth as a reference to correct the entire image. The geometric correction was undertaken by automatically matching the data of each frame of the hyperspectral imaging sensor and the corresponding posture information, which was recorded in real time through the IMU/GPU module.

[1]https://drive.google.com/drive/folders/1HMTIodCUcXgMIaijOco_wC8uKjw6cSnl?usp=sharing

C. Database

The YRD database contains three HSIs, i.e., the NC12 dataset, the NC16 dataset, and the NC13 dataset. On September 23, 2020, the NC12 and NC16 datasets were acquired with clear and cloudless weather. We acquired the NC12 dataset from 12:27 to 13:10; the spatial resolution of the NC12 dataset is about 0.182 m because the flying height of the UAV is 300 m. There are 12 types of land covers. The NC16 dataset is acquired from 13:50 to 14:20, and the spatial resolution of the NC16 dataset is about 0.266 m because the flying height of the UAV is 400 m. There are 16 types of land covers. In addition, the NC13 dataset was acquired from 14:47 to 15:20 on September 24, 2020, with a cloudy day. The flying height of the UAV is 300 m, and the spatial resolution of the NC13 dataset is about 0.182 m. Figs. 2–4 show the created datasets.

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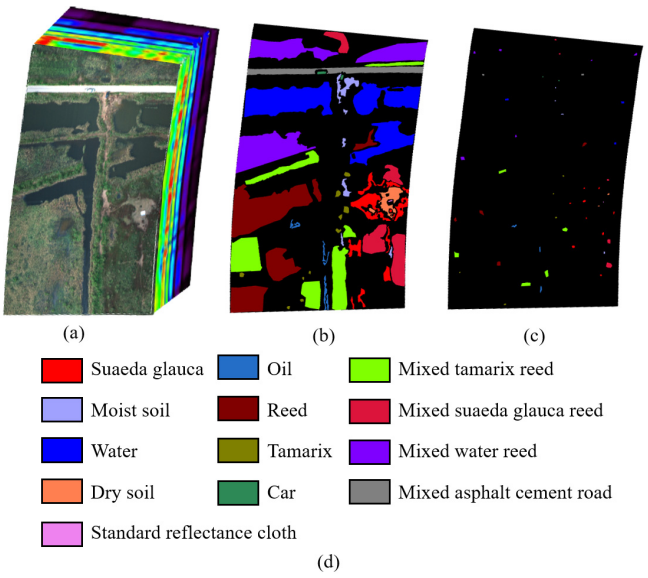


Fig. 4. NC13 dataset. (a) Image cube. (b) Ground-truth image. (c) Training data visualization. (d) Class name.

III. Methodology

In this section, we briefly introduce the MGSSAN framework. Then, the basic structure of CNNs is introduced. Next, the GSSA used for spectral–spatial feature fusion is described in detail. Finally, the ERW optimization step is introduced.

A. MGSSAN Framework

Fig. 5 shows the flowchart of the MGSSAN framework. The framework consists of three parts: the basic structure of CNNs, an attention mechanism named GSSA, and extended random walk-based optimization. In order to better obtain spectral and spatial information, we utilize two-branch CNNs to extract spectral–spatial features, respectively. First, for a given pixel, we select $11 \times 11 \times B$ cube with the pixel as the center pixel, where B is the number of HSI bands, and then, the small cube is fed into two-branch CNNs. Second, we utilize the GSSA mechanism to generate GSSA for each spectral and spatial layer, so as to stimulate the two branches to focus more on significant information while suppressing unimportant information. Finally, we use a fully connected layer to yield the classification probabilities map followed by ERW to obtain the final classification map.

In order to obtain the best parameters for classification, we divide the labeled data into three parts, including the training set, the test set, and the validation set. The data in the training set are used to update the network parameters, and the validation set can monitor the training process to obtain the optimal parameters. The test set is used to evaluate the generalization performance of the method. For parameter updates, the cross-entropy loss is adopted to update the parameters, which can be defined as

$$\text{Loss} = -\frac{1}{C}\sum_{i=0}^{C}\left(y_i \log\left(\hat{y_i}\right) + (1-y_i)\log\left(1-\hat{y_i}\right)\right) \quad (2)$$

where C represents the category number, and y_i and $\hat{y_i}$ represent the ground-truth and predicted labels, respectively.

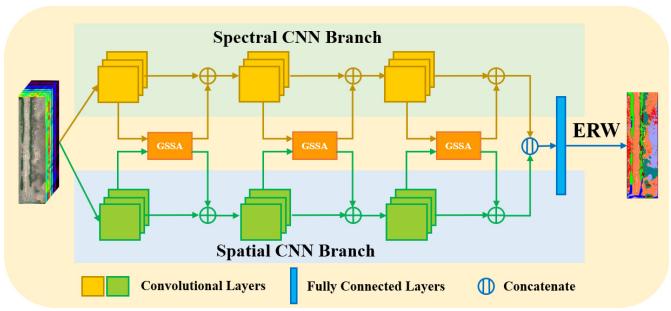


Fig. 5. Overview of the proposed MGSSAN framework for mapping coastal wetlands.

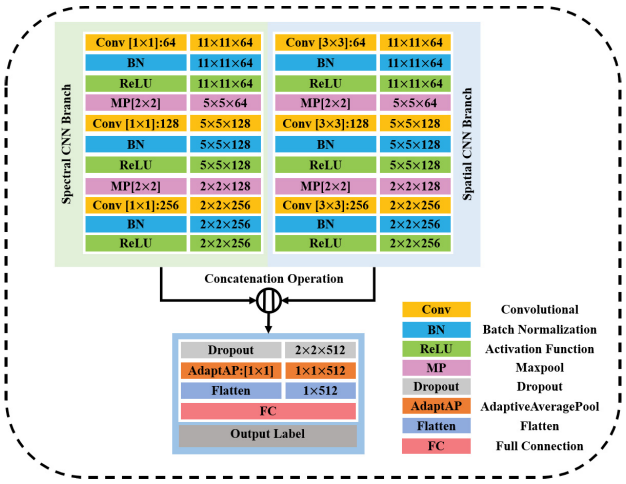


Fig. 6. Overall parameters configuration of the basic two-branch CNN.

B. Basic Structure of CNNs

As described in the MGSSAN framework part, the two-branch CNNs are adopted for the proposed MGSSAN method. In this section, we will elaborate on the two-branch networks.

For HSI classification, it is difficult to train very deep CNNs with limited training samples. We introduce a basic two-branch CNNs, in which three convolutional layers are applied to construct the basic structure of CNN for spectral and spatial branches. To stabilize and accelerate the training process, a batch normalization layer is applied after the convolutional layer to improve the abilities of nonlinear representation of the network, and a rectified linear unit (ReLU) is followed by the batch normalization layer. Before the second and third convolutional layers, a max-pooling layer is adopted for reducing model parameters and computational complexity. Meanwhile, dropout is utilized to prevent overfitting. Fig. 6 elaborates on more details about the basic CNN. The spectral branch uses a small kernel, such as 1×1, to make the spectral branch focus more discriminative channels while suppressing unnecessary ones. On the contrary, the spatial branch pays more attention to the semantic positions by a bigger kernel, such as 3×3. The number of filters ranges from 64 to 256 for detailed feature extraction. The bth branch feature map of the lth convolutional layers can be described as

$$\mathbf{F}_b^l = \delta\left(\mathbf{F}_b^{l-1} * \mathbf{w}_b^l + k_b^l\right) \quad (3)$$

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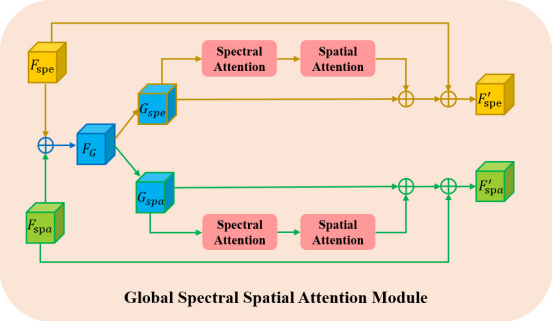


Fig. 7. Architecture of the global spectral–spatial attention module.

where $b \in \{1, 2\}$, $l \in \{1, 2, 3\}$, $\mathbf{F}_b^l$ represents the features of the lth convolutional layer from the bth branch, and w is the convolutional kernel whose size is alterable in different feature extraction branches. Specifically, $\mathbf{w}_b^l \in \mathbb{R}^{1\times 1}$ when it is in the spectral branch, and $\mathbf{w}_b^l \in \mathbb{R}^{3\times 3}$ when it is in the spatial branch. "*" is the convolutional operator, k_b^l is the bias, and δ is the ReLU activation function.

In addition, residual structures are adopted to combine features of different convolutional layers, which makes the network become much easier to learn and faster to train without any additional parameters. Moreover, we apply the attention module for each layer to make full use of the convolutional features of each layer. For the effective aggregation of spectral and spatial information, we concatenate the features extracted by the spectral and spatial branches.

C. GSSA

In this section, the proposed GSSA mechanism is depicted detailedly. As shown in Fig. 7, GSSA generates a global joint representation $\mathbf{F}_G$ of spectral–spatial features by fusing the spectral features $\mathbf{F}_{\text{spe}}$ and the spatial features $\mathbf{F}_{\text{spa}}$, which are obtained from the spectral CNN branch and spatial CNN branch, and the global features $\mathbf{F}_G$ could be utilized to adaptively guide the basic CNNs to focus on different spectral channels and spatial locations by multiplying the attention weights. Therefore, it could perform on each channel and each spatial pixel, so as to highlight the useful features and suppress unimportant information. Finally, after the process of GSSA, more effective spectral features $\mathbf{F}'_{\text{spe}}$ and spatial features $\mathbf{F}'_{\text{spa}}$ can be obtained, respectively.

Specifically, the global spectral–spatial features $\mathbf{F}_G$ are used to generate attention weights for spectral and spatial feature extractions, respectively, which $\mathbf{F}_G$ can be defined as

$$\mathbf{F}_G = \mathbf{F}_{\text{spe}} + \mathbf{F}_{\text{spa}} \tag{4}$$

where $\mathbf{F}_G \in \mathbb{R}^{B\times H\times W}$ is aggregated features and contains both spectral information and spatial context. In order to update the basic CNNs parameters more efficiently, we adopt two convolutional layers to divide the global features into two parts, so as to improve the generalization of the global features. One part of the features is named $\mathbf{G}_{\text{spe}}$, which has spectral sensitivity, the other features are named $\mathbf{G}_{\text{spa}}$, which has spatial sensitivity. This could be calculated as follows:

$$\begin{cases} \mathbf{G}_{\text{spe}} = \text{conv}_{\text{spe}}(\mathbf{F}_G) \\ \mathbf{G}_{\text{spa}} = \text{conv}_{\text{spa}}(\mathbf{F}_G) \end{cases} \tag{5}$$

where conv_{spe} and conv_{spa} represent two different convolution operations with 3×3 kernel, and they are used to divided the global features into a spectral sensitivity part G_{spe} and a spatial sensitivity part G_{spa}. Then, we utilize a series of spectral and spatial attention modules to obtain a weight map that could improve spectral separability and learn the spatial context details. We take the global features with spectral sensitive $\mathbf{G}_{\text{spe}}$ as an input feature; the overall process about the attention module could be summarized as follows:

$$\begin{aligned} \mathbf{F}_{G_1} &= \mathbf{M}_{\text{spe}}\left(\mathbf{G}_{\text{spe}}\right) \otimes \mathbf{G}_{\text{spe}} \\ \mathbf{F}_{G_2} &= \mathbf{M}_{\text{spa}}(\mathbf{F}_{G_1}) \otimes \mathbf{F}_{G_1} \end{aligned} \tag{6}$$

where $\mathbf{F}_{G_1}$ represents the spectral attention weight, $\mathbf{F}_{G_2}$ represents the spatial attention weight, $\mathbf{M}_{\text{spe}}$ and $\mathbf{M}_{\text{spa}}$ stand for the 1-D spectral attention mechanism and the 2-D spatial attention mechanism, respectively, and $\otimes$ denotes elementwise multiplication. Afterward, we will introduce each attention module in detail.

In particular, for the spectral attention mechanism, we obtain an average pooling (AvgPool) feature $\mathbf{F}_{\text{avg}}^{\text{spe}}$ and a maximum pooling (MaxPool) feature $\mathbf{F}_{\text{max}}^{\text{spe}}$ by utilizing global AvgPool and global MaxPool operations on the input features, separately, and then, the two descriptors are fed into a multilayer perceptron (MLP) with a hidden layer and shared weights. In order to reduce the computational complexity of the model, the hidden activation size is set to $\mathbb{R}^{B/r\times 1\times 1}$, where r denotes the reduction rate. Finally, an elementwise summation is used for the outputs of MLP, which is shown as

$$\begin{cases} \mathbf{F}_{\text{avg}}^{\text{spe}} = \text{AvgPool}\left(\mathbf{G}_{\text{spe}}\right) \\ \mathbf{F}_{\text{max}}^{\text{spe}} = \text{MaxPool}\left(\mathbf{G}_{\text{spe}}\right) \\ \text{MLP}(\cdot) = \mathbf{W}_1(\delta(\mathbf{W}_0(\cdot))) \end{cases} \tag{7}$$

$$\mathbf{M}_{\text{spe}} = \sigma\left(\text{MLP}\left(\mathbf{F}_{\text{avg}}^{\text{spe}}\right) + \text{MLP}\left(\mathbf{F}_{\text{max}}^{\text{spe}}\right)\right) \tag{8}$$

where σ denotes the sigmoid function, $\mathbf{W}_0 \in \mathbb{R}^{B/r\times B}$, and $\mathbf{W}_1 \in \mathbb{R}^{B\times B/r}$. The spectral attention feature maps are obtained via multiplying the weight by the input features, which is valuable to give the response on the channels with a powerful discriminative ability.

In addition, the spatial attention module is a powerful method to adaptively improve the feature expression in critical spatial locations, so as to make the network achieve a better classification performance. Thus, to implement this module, we utilize two pooling operations on the input image to obtain two channels feature map, including AvgPool features $\mathbf{F}_{\text{avg}}^{\text{spa}} \in \mathbb{R}^{1\times H\times W}$ and max-pooling features $\mathbf{F}_{\text{max}}^{\text{spa}} \in \mathbb{R}^{1\times H\times W}$, and then, the two-channel feature maps are convolved by a standard convolutional layer to produce the spatial attention map. The spatial attention mechanism can be expressed as

$$\begin{cases} \mathbf{F}_{\text{avg}}^{\text{spa}} = \text{AvgPool}(\mathbf{F}_{G_1}) \\ \mathbf{F}_{\text{max}}^{\text{spa}} = \text{MaxPool}(\mathbf{F}_{G_1}) \end{cases} \tag{9}$$

$$\mathbf{M}_{\text{spa}} = \sigma\left(f^{3\times 3}\left(\left[\mathbf{F}_{\text{avg}}^{\text{spa}}; \mathbf{F}_{\text{max}}^{\text{spa}}\right]\right)\right) \tag{10}$$

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TABLE I
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS FOR THE NC12 DATASET

Class Name	Training	Test	Methods							
			1DCNN	CDCNN	3DCNN	SSRN	DBMA	DBDA	HybridSN	Our method
Suaeda glauca	4467	9650	99.55	99.82	99.79	**99.98**	99.77	99.59	99.79	99.47
Reed	531	15072	65.53	69.20	75.91	**76.08**	63.58	70.23	74.16	70.38
Water	575	35716	97.24	97.10	99.76	98.93	99.50	99.90	96.03	**100.00**
White cloth	25	403	66.25	83.13	92.80	74.69	62.03	68.49	52.85	**94.79**
Withered reed	1543	59741	88.88	95.68	88.87	95.44	89.11	93.92	92.31	**96.99**
Tamarix	2381	6876	95.80	88.14	94.58	96.73	97.69	97.31	96.89	**98.97**
Stone	121	1046	83.92	74.22	90.49	**96.00**	93.43	91.72	61.66	92.20
Moist soil	1222	67275	99.92	96.60	98.27	95.26	94.96	98.27	91.85	**99.98**
Mudflat	123	511	76.35	81.92	92.12	75.38	**97.88**	86.54	50.58	87.12
Spartina alterniflora	1373	122947	85.37	86.73	88.03	95.75	**96.44**	92.30	85.04	94.51
Dry soil	2032	118041	89.86	98.38	98.10	97.95	98.38	98.23	95.61	**98.98**
Standard reflectance cloth	9	66	**100.00**	56.06	77.27	93.94	**100.00**	**100.00**	65.15	**100.00**
OA			90.03	93.13	93.36	95.88	94.92	95.12	90.85	**96.68**
AA			87.39	85.58	91.33	91.35	91.07	91.38	80.16	**94.45**
Kappa			87.72	91.49	91.80	94.88	93.67	93.95	88.72	**95.86**

where $f^{3\times3}$ represents a convolution operation that the filter size is 3 × 3. Then, the spectral–spatial attention feature maps of spectral CNN branch $\mathbf{F}''_{\text{spe}}$ could be obtained by multiplying the spectral attention feature maps and spatial attention weight. Finally, to avoid overfitting, we refer to the residual connections and add the input feature $\mathbf{F}_{\text{spe}}$ from the spectral CNN branch and the spectral–spatial attention feature maps $\mathbf{F}''_{\text{spe}}$ together to generate the output feature $\mathbf{F}'_{\text{spe}}$ of the GSSA module for the spectral CNN branch. Similarly, the output feature $\mathbf{F}'_{\text{spa}}$ of the GSSA module for the spatial CNN branch can be obtained with the same operation.

In summary, spectral attention and spatial attention are effective to promote feature extraction for HSI classification. With the GSSA module, the important information would be highlighted, and the insignificant information would be suppressed.

D. Extended Random Walker Optimization

The classification map estimated by the two-branch CNNs is usually noisy, which usually reduces classification performance. ERW [49] is a widely used probabilistic optimization model, which is to model the spatial correlation among adjacent pixels. In this work, the ERW is adopted to optimize the initial classification probabilities, which can well remove the noisy labels in the classification map. In this situation, the boundaries of different ground covers are well aligned with the edges of different objects. Specifically, the initial classification probabilities map $\mathbf{M}$ obtained by the dual-path networks can be optimized by minimizing the following energy function:

$$E^n(\mathbf{P}_n) = E^n_{\text{spatial}}(\mathbf{P}_n) + \gamma\, E^n_{\text{aspatial}}(\mathbf{P}_n) \tag{11}$$

where $\mathbf{P}_n \in \mathbb{R}^{H\times W}$ is the class probability for the nth category that can be obtained by minimizing the energy function, n refers to the class number, and γ is a free parameter. The energy function has two major parts: one is the spatial term, and the other is the aspatial term. In addition, the spatial relativity among contiguous pixels can be modeled well by the spatial term E^n_{spatial}, which can be defined as

$$E^n_{\text{spatial}}(\mathbf{P}_n) = \mathbf{P}_n^T \mathbf{L} \mathbf{P}_n \tag{12}$$

where $\mathbf{L}$ indicates the Laplacian matrix of a weight graph

$$\mathbf{L}_{ij} = \begin{cases} \sum e^{-\beta\left(\upsilon_i-\upsilon_j\right)^2}, & \text{if } i = j \\ -e^{-\beta\left(\upsilon_i-\upsilon_j\right)^2}, & \text{if } i \text{ and } j \text{ are adjacent pixels} \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where υ_i stands for the ith pixel value and β is a free parameter. The other term E^n_{aspatial} is given as follows:

$$E^n_{\text{aspatial}}(\mathbf{P}_n) = \sum_{q=1,q\neq n} \mathbf{P}_q^T \Lambda_q \mathbf{P}_q + (\mathbf{P}_n - 1)^T \Lambda_n (\mathbf{P}_n - 1) \tag{14}$$

where Λ_b denotes a diagonal matrix, where diagonal values is made up of the initial probability map $\mathbf{M}$. Finally, the optimized probability map $\mathbf{M}'$ can be obtained by calculating the following equation:

$$\mathbf{M}' = \arg\max_n \mathbf{P}_n. \tag{15}$$

IV. EXPERIMENTS AND ANALYSIS

In this section, three YRD datasets are used to validate the effectiveness of the proposed MGSSAN framework. All the deep learning models are simulated by PyTorch on Python3.6 running on a personal computer equipped with Windows 10 and NVIDIA GeForce RTX 2080 Ti.

A. Experiments Setting

1) Datasets: In the experiments, three hyperspectral wetland images are utilized to evaluate the performance of the proposed framework, and all three datasets are collected by field survey. For each dataset, a small number of samples are selected as training samples, and the rest is used as the test set. The number of training samples of NC12, NC16, and NC 13 is set as 3.19%, 2.81%, and 1.63%, respectively. The detailed number of each class is shown in the second and third columns of Tables I–III.

2) Parameters Setting: All networks in this article are implemented using the PyTorch platform, and we take 11 × 11 neighborhood samples as the input size except 1DCNN [50], which take 1 × 1 neighborhood samples as the input size. In addition, we use Adam [51] to optimize the networks, and the initial learning rate is set to 0.005 except that

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TABLE II
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS FOR THE NC16 DATASET

Class Name	Training	Test	Methods							
			1DCNN	CDCNN	3DCNN	SSRN	DBMA	DBDA	HybridSN	Our method
Suaeda glauca	1325	9209	99.97	99.58	99.89	99.52	99.32	99.39	99.77	**99.99**
Cement road	74	1461	**97.74**	84.53	83.30	76.45	69.61	94.39	78.23	87.75
Asphalt road	141	4300	80.68	79.34	**91.77**	81.17	76.78	89.10	76.24	91.14
Water	434	69638	98.71	99.25	99.52	97.77	96.29	99.07	94.96	**99.76**
Stone	273	5244	83.82	83.74	90.59	87.74	73.84	88.77	**92.56**	90.33
Hay	58	2374	75.86	71.31	61.29	71.78	70.05	77.46	65.92	**80.83**
Iron	2	32	0	0	15.63	40.63	**100.00**	90.63	31.25	**100.00**
Tamarix	61	559	59.03	80.68	55.46	49.73	94.45	97.32	18.25	**97.50**
Withered reed	96	1810	57.90	51.71	57.57	78.45	64.64	59.72	**87.51**	54.25
Reed	57	1187	57.20	**85.00**	44.14	77.84	82.39	82.14	72.37	80.29
Spartina alterniflora	459	4839	96.94	92.90	93.51	92.63	95.24	96.08	85.76	**98.04**
Moss	29	704	**80.40**	24.57	52.70	61.65	68.18	57.67	29.83	76.42
Dry soil	705	18775	93.30	88.19	**95.49**	89.95	91.87	89.72	93.06	92.91
Moist soil	554	28822	93.75	94.61	92.21	**98.52**	96.77	95.01	97.01	97.14
Mudflat	71	1051	89.06	84.49	77.64	93.34	**98.57**	96.57	78.78	98.19
Standard reflectance cloth	4	32	**100.00**	68.75	50.00	37.50	**100.00**	**100.00**	93.75	**100.00**
OA			94.55	93.85	94.64	94.64	93.35	95.18	92.88	**96.53**
AA			79.02	74.29	72.54	77.17	86.13	88.31	74.70	**90.28**
Kappa			92.51	91.53	92.61	92.62	90.92	93.37	90.29	**95.21**

TABLE III
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS FOR THE NC13 DATASET

Class Name	Training	Test	Methods							
			1DCNN	CDCNN	3DCNN	SSRN	DBMA	DBDA	HybridSN	Our method
Suaeda glauca	400	14225	86.22	84.90	84.48	90.54	93.37	97.52	82.22	**99.35**
Mixed asphalt cement road	108	13184	98.52	86.98	98.96	99.07	95.37	74.89	94.56	**99.74**
Moist soil	98	4910	69.08	62.28	75.32	91.55	90.41	**92.40**	70.45	80.63
Water	230	63007	99.57	95.89	98.08	99.87	97.63	99.71	91.14	**99.87**
Oil	309	1963	95.32	93.82	99.20	97.69	**100.00**	97.84	99.20	96.33
Reed	665	56922	49.76	51.29	53.08	53.04	48.00	56.99	59.57	**69.29**
Tamarix	134	1954	46.93	70.62	64.33	70.83	73.80	70.06	57.37	**77.12**
Car	31	391	70.66	76.79	48.98	**84.95**	60.20	81.89	37.76	79.34
Dry soil	54	4427	84.23	81.00	65.26	**87.76**	64.13	64.85	76.69	81.55
Standard reflectance cloth	8	49	92.31	25.00	59.62	**100.00**	**100.00**	**100.00**	61.54	**100.00**
Mixed tamarix reed	1465	39837	56.00	57.37	53.04	48.30	56.30	**57.98**	53.64	47.92
Mixed suaeda glauca reed	638	21514	73.40	66.52	74.34	74.57	77.26	**77.67**	76.47	68.99
Mixed water reed	435	53549	68.55	73.14	71.22	72.26	**79.67**	79.24	78.46	76.82
OA			72.96	72.35	73.32	74.31	75.14	76.93	74.43	**78.27**
AA			76.19	71.20	72.76	82.34	79.70	80.85	72.24	**82.84**
Kappa			67.90	67.24	68.31	69.54	70.58	72.61	69.72	**74.21**

CDCNN [52] is 0.0005 and linearly decays for every 20 iterations. Besides, the number of maximum training epochs is set to 100.

3) Evaluation Indices: The overall accuracy (OA), average accuracy (AA), and kappa coefficient are used to quantitatively evaluate the performance of different classification methods in the work. OA represents the ratio of all correctly classified samples in the total labeled samples. AA is defined as the average value on the percentages of each category being correctly classified. The kappa coefficient is an index used for consistency testing.

B. Comparison Methods

In order to verify the superiority of the proposed method on the YRD dataset, we compared it with seven state-of-the-art classification methods, including 1DCNN [50], CDCNN [52], 3DCNN [36], SSRN [53], DBMA [44], DBDA [46], and HybridSN [37].

1) *1DCNN:* The 1DCNN architecture contains five layers with weight: the input layer, the 1-D convolutional layer, the max-pooling layer, the full connection layer, and the output layer. 1DCNN is a single-branch spectral classification method, which only considers the spectral information and ignores the spatial information.
2) *CDCNN:* The CDCNN method utilizes a multiscale convolutional filter bank to achieve the joint exploitation of the spectral and spatial information and introduces a higher and deeper network with the help of "residual learning."
3) *3DCNN:* The 3DCNN method utilizes 3-D convolutions to jointly extract spectral and spatial features.
4) *SSRN:* The SSRN method introduces residual blocks to connect every other 3-D convolutional layer through identity mapping, which can further improve HSI classification performance.
5) *DBMA:* The DBMA is a two-branch spectral–spatial classification method, which adds dense connection and an attention mechanism on the basis of 3DCNN.
6) *DBDA:* The DBDA is also a two-branch spectral–spatial classification method that is inspired by DBMA, different from the framework of DBMA; DBDA introduces different attention mechanisms in the spectral extraction and spatial feature extraction branches.

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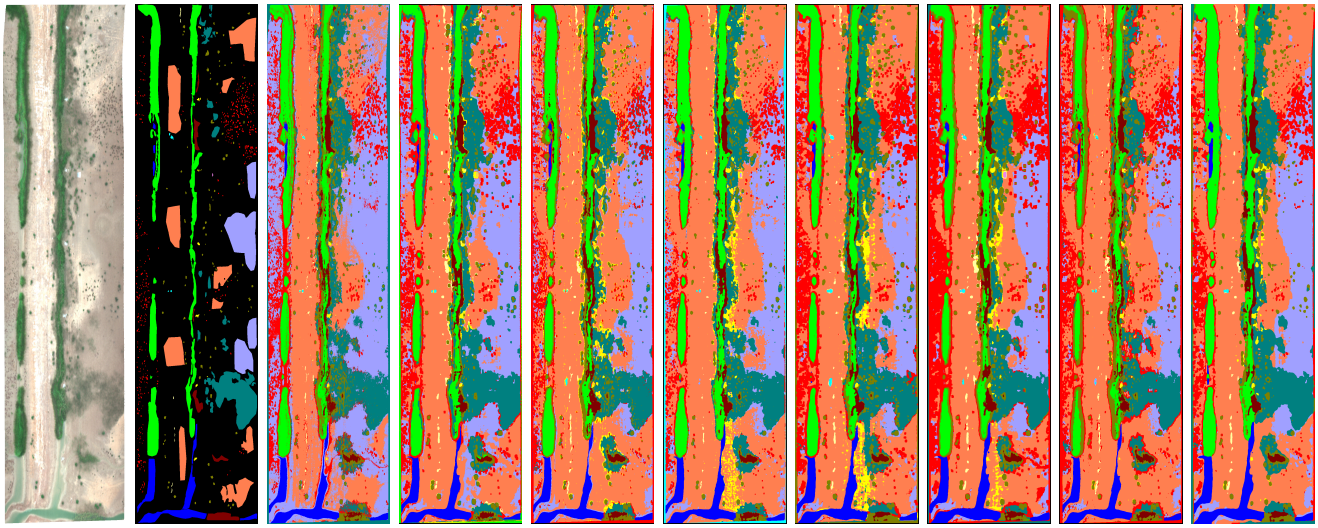


Fig. 8. Classification maps obtained by different methods on the NC12 dataset. (a) False-color image. (b) Ground-truth image. (c) 1DCNN, OA = 90.03%. (d) CDCNN, OA = 93.13%. (e) 3DCNN, OA = 93.36%. (f) SSRN, OA = 95.88%. (g) DBMA, OA = 94.92%. (h) DBDA, OA = 95.12%. (i) HybridSN, OA = 90.85%. (j) Our method, OA = 96.68%.

7) *HybridSN:* The HybridSN utilizes 3-D convolutions and 2-D convolutions to jointly extract spectral and spatial features.

Table I shows the objective accuracies of different methods on the NC12 dataset. It can be clearly observed that the proposed method obtains the highest accuracies in terms of OA, AA, and kappa coefficient. Moreover, the proposed method yields the best classification performance on several classes. Fig. 8 shows the classification maps of different methods. As shown in this figure, 1DCNN obtains a noisy classification map. The reason is that 1DCNN cannot effectively extract spatial information, and thus, the classification result is unsatisfactory. CDCNN and HybridSN methods employ spatial information over 1DCNN. The classification performance of the two methods has been improved. Although 3-D convolution can better utilize the spatial–spectral information of different land covers, it contains more training parameters than CDCNN. To solve this problem, DBDA, DBMA, and SSRN methods introduce the attention mechanism and residual learning into 3DCNN, which further improves classification performance. The proposed method produces the best classification performance compared to other classification methods. The main reason is that the GSSA mechanism can effectively characterize the spatial–spectral features of different objects.

Table II lists the objective metrics of all considered methods, and Fig. 9 shows the classification maps obtained by different methods on the NC16 dataset. The 1DCNN and CDCNN methods cannot identify the "Iron" class because the number of training samples is very scarce. By introducing spectral convolution, the 3DCNN method achieves better classification performance compared with 1DCNN and CDCNN methods. For the DBDA and DBMA methods, they cannot well classify the river in the classification map. By contrast, the proposed method can achieve the best performance among all compared approaches on the NC16 dataset since we make full use of the attention mechanism guided by global information.

From the first column of Table III, it is obvious that there have some mixed land covers on the NC13 dataset, which makes the classification task very difficult. Table III shows the corresponding quantitative results of different methods on the NC13 dataset. Fig. 10 shows the classification maps of all approaches. 1DCNN, CDCNN, and 3DCNN cannot perform well on the NC13 dataset. This is due to the fact that their feature extractors fail to extract the discriminative features under complex scenes. SSRN improves the classification performance by introducing residual learning. HybridSN combines 2DCNN and 3DCNN to improve the classification result. However, there are some misclassified pixels in some objects. Different from other approaches, the proposed method obtains the best classification result in terms of both objective and subjective performance. Furthermore, it is found that attention-based methods, such as DBMA, DBDA, and the proposed method obtain better classification performance, which also demonstrates that the attention mechanism is indeed a good technique to improve classification accuracy. Among all attention-based approaches, our GSSA module obtains the highest classification accuracies by integrating spectral and spatial information.

C. Discussion

Effects of Different Attention Modules: In order to verify the effectiveness of the proposed GSSA module, two widely used attention modules are adopted for comparison, such as the squeeze-and-excitation (SE) attention module [54] and the convolutional block attention module (CBAM) [42]. Furthermore, we also analyze the influence of the proposed method without the GSSA module. Table IV presents the classification results of the proposed method with different attention modules on the YRD datasets. The proposed method with the GSSA module achieves the best classification performance with respect to other attention modules, which also verifies the effectiveness of the proposed attention module. The main

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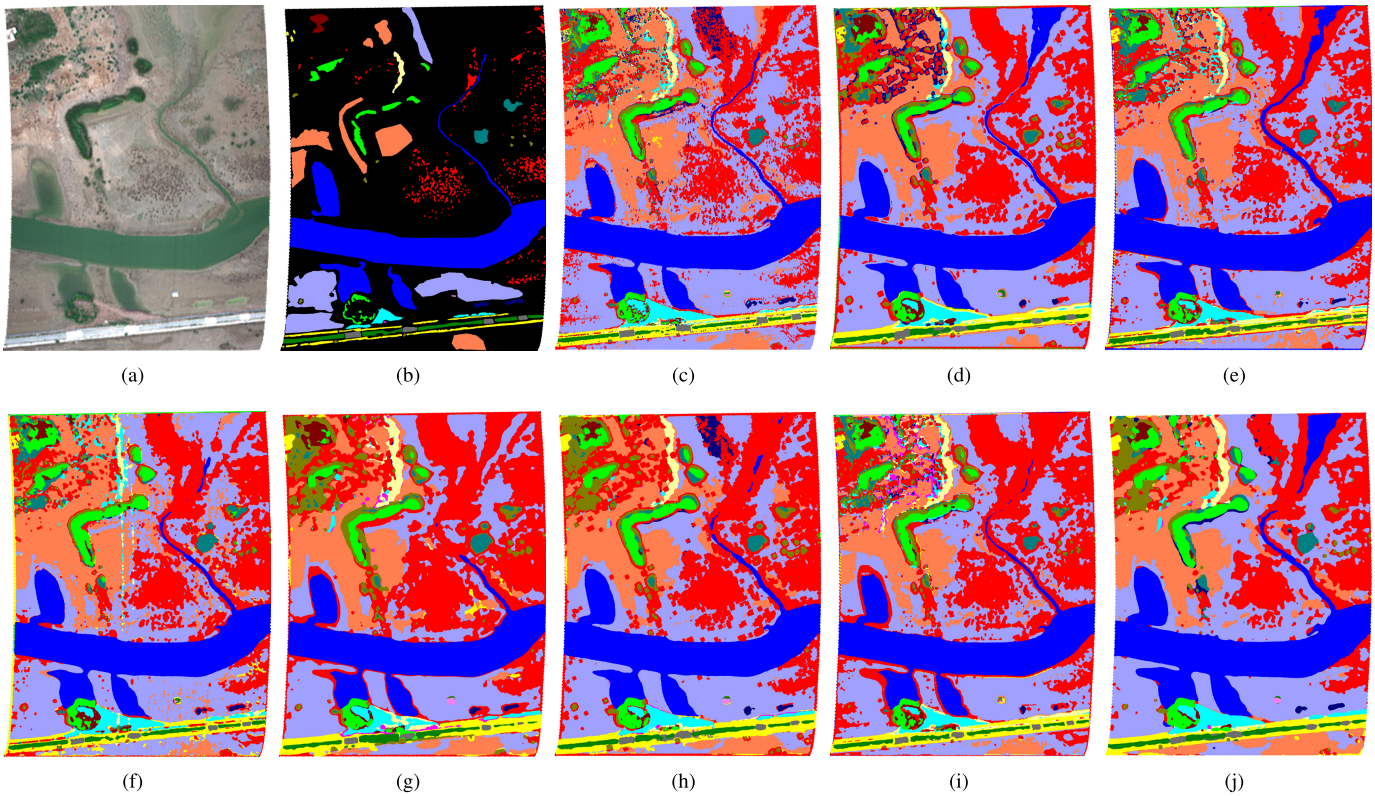


Fig. 9. Classification maps obtained by different methods on the NC16 dataset. (a) False-color image. (b) Ground-truth image. (c) 1DCNN, OA = 94.55%. (d) CDCNN, OA = 93.85%. (e) 3DCNN, OA = 94.64%. (f) SSRN, OA = 94.64%. (g) DBMA, OA = 93.35%. (h) DBDA, OA = 95.18%. (i) HybridSN, OA = 92.88%. (j) Our method, OA = 96.53%.

TABLE IV
ABLATION ANALYSIS FOR DIFFERENT ATTENTION MODULES ON THREE DATASETS

		TBCNN	SE+TBCNN	CBAM+TBCNN	Our method
NC12	OA	96.19	96.41	96.47	**96.68**
	AA	**95.44**	94.23	94.41	94.45
	Kappa	95.25	95.52	95.60	**95.86**
NC16	OA	96.16	96.24	96.30	**96.53**
	AA	88.68	89.68	90.20	**90.28**
	Kappa	94.70	94.81	94.89	**95.21**
NC13	OA	76.80	77.18	77.70	**78.27**
	AA	81.83	82.67	80.29	**82.84**
	Kappa	72.46	72.91	73.48	**74.21**

TABLE V
ABLATION ANALYSIS FOR ERW ON THREE DATASETS

		Without ERW	With ERW
NC12	OA	96.29	**96.68**
	AA	94.44	**94.45**
	Kappa	95.40	**95.86**
NC16	OA	96.45	**96.53**
	AA	89.10	**90.28**
	Kappa	95.10	**95.21**
NC13	OA	77.39	**78.27**
	AA	81.97	**82.84**
	Kappa	73.17	**74.21**

reason is that the GSSA module could not only focus on the important spectral information but also mark the necessary spatial location.

Analyzing the Influence of the ERW: To illustrate the classification performance of the proposed method with or without the ERW, an experiment is performed on three datasets. Table V shows the classification performance of the proposed method with or without the ERW model on all considered datasets. It is easily observed that the proposed method with the ERW model yields better classification performance in terms of OA, AA, and kappa coefficient. This experiment also verifies that the ERW is helpful for improving the classification performance by utilizing the adjacent spatial information.

Analyzing the Influence of Different Spatial Sizes: In order to discuss the sensitivity of the input spatial size on different datasets, an experiment is performed on the created database, in which the spatial size ranges from 7×7 to 15×15. Table VI lists the classification accuracies of the proposed method with different spatial sizes. It can be seen that the classification performance obtained by the proposed method first tends to increase and then gets worse when the spatial size is gradually increasing. The reason is that, when the spatial size is relatively large, the pixels in the local region belong to different types of objects. In this situation, the classification performance of the proposed method tends to decrease. Based on the classification accuracies in Table VI, the spatial size is set to be 11×11 for all used datasets.

Analyzing the influence of different values of β and γ: For the ERW method, the influence of parameters, i.e., β and γ, to the proposed method is analyzed. Fig. 11 shows the classification performance of the proposed method when the two parameters β and γ are varying. When the value of γ is fixed,

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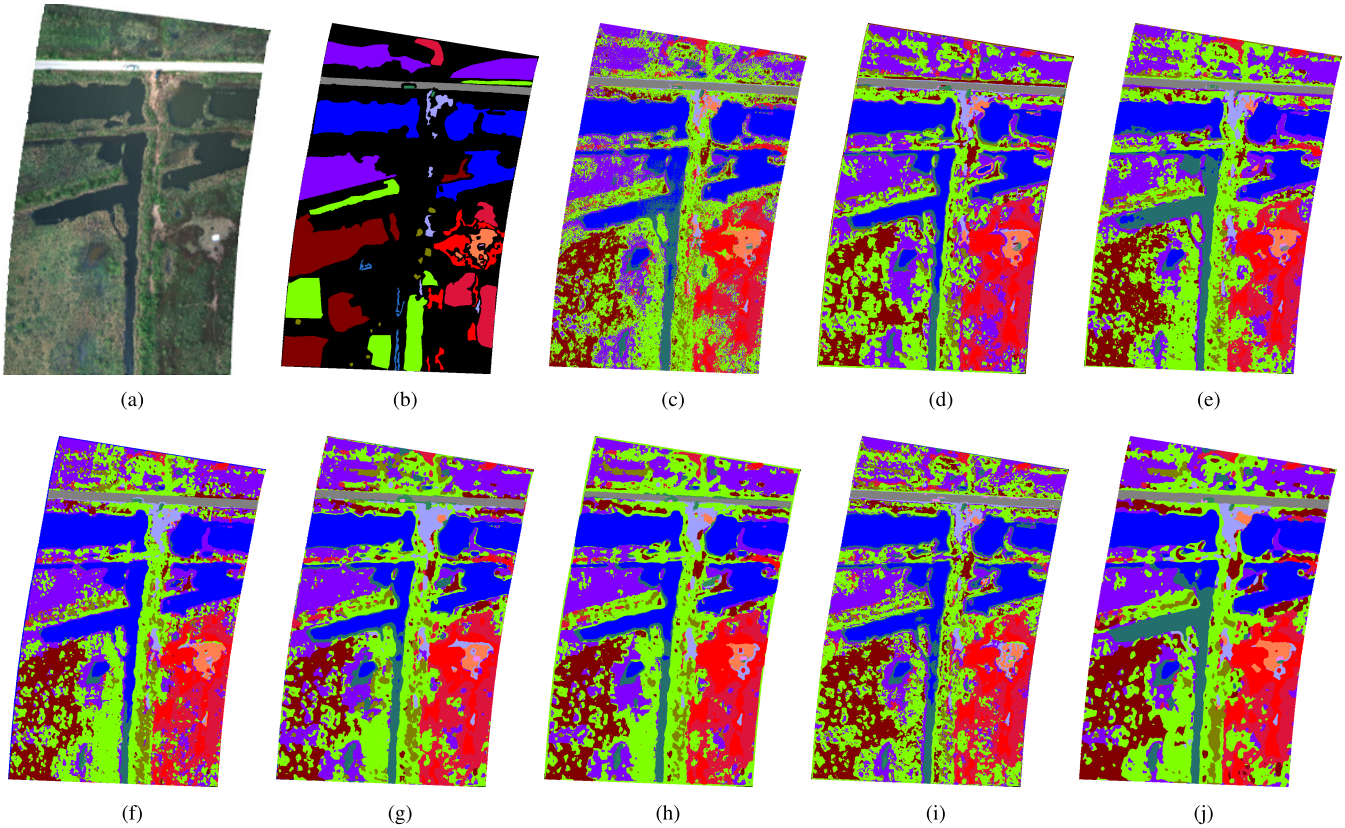


Fig. 10. Classification maps obtained by different methods on the NC13 dataset. (a) False-color image. (b) Ground-truth image. (c) 1DCNN, OA = 72.96%. (d) CDCNN, OA = 72.35%. (e) 3DCNN, OA = 73.32%. (f) SSRN, OA = 74.31%. (g) DBMA, OA = 75.14%. (h) DBDA, OA = 76.93%. (i) HybridSN, OA = 74.43%. (j) Our method, OA = 78.27%.

TABLE VI

ABLATION ANALYSIS FOR DIFFERENT INPUT SPATIAL SIZES ON THREE DATASETS

Input spatial size		7*7	9*9	11*11	13*13	15*15
NC12	OA	95.67	96.58	96.68	**97.09**	97.00
	AA	93.21	94.34	94.45	**95.36**	94.10
	Kappa	94.61	95.74	95.86	**96.36**	96.26
NC16	OA	96.19	96.11	**96.53**	95.03	95.98
	AA	89.02	91.11	**90.28**	90.52	90.16
	Kappa	94.75	94.64	**95.21**	93.16	94.44
NC13	OA	75.07	76.27	**78.27**	76.78	77.65
	AA	81.36	82.13	**82.84**	82.24	80.61
	Kappa	70.46	71.84	**74.21**	72.45	73.52

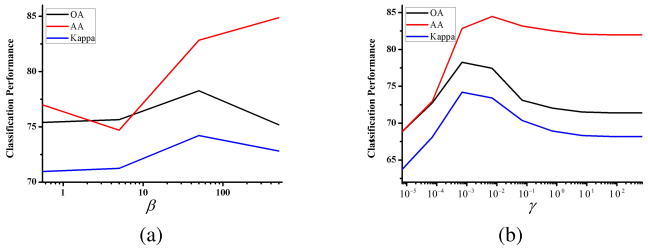


Fig. 11. Sensitivity analysis for the free parameters β and γ for the NC13 dataset. (a) Influence of the parameters on classification performance with different values of β, while the other parameter is $\gamma = 0.0007$. (b) Influence of the parameters on classification performance with different values of γ, while the other parameter is $\beta = 50$.

Fig. 11(a) shows the influence of β to the proposed method. It can be seen that the classification performance of the proposed method tends to decrease when β is relatively small. The reason is that the spatial correlation between adjacent pixels cannot be well modeled by the edge weights. When β is increasing, the classification accuracies of the proposed method tend to decrease. This is due to the fact that the balance of the spatial and aspatial energy functions cannot be well preserved. Similarly, when we discuss the parameter γ, the value of β is fixed. It is obvious from Fig. 11(b) that the classification performance of the proposed method first increases and then tends to decrease. Based on this sensitivity analysis, $\beta = 50$ and $\gamma = 7 \times 10^{-3}$ are set as the default parameters for the following experiments.

Analyzing the Influence of the Dropout Rate: The effect of different dropout rates, i.e., d, is discussed. The experiment is performed on the YRD dataset. d is ranging from 0.1 to 0.9. Table VII gives the classification accuracies of the proposed method with different dropout rates. It is obvious that the proposed method obtains the best classification performance on the NC13 and NC16 datasets when the dropout rate is 0.5. Furthermore, when the dropout rate is 0.4, the proposed method achieves the best classification performance on the NC12 dataset. Thus, from Table VII, it is clear to observe that MGSSAN increases the performance when d increases, and then, the classification performance of the proposed method tends to decrease. Therefore, d is set to 0.4, 0.5, and 0.5 for NC12, NC16, and NC13, respectively.

Analyzing the Computing Time of Different Methods: Table VIII records the training time and test time of different methods. In Table VIII, it is clear to observe that 1DCNN

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TABLE VII

ABLATION ANALYSIS FOR DIFFERENT DROPOUT RATES ON THREE DATASETS

Dropout rate		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
NC12	OA	96.20	96.59	96.43	**96.68**	96.50	95.27	96.42	96.53	95.77
	AA	93.98	94.67	95.38	**94.45**	94.20	93.85	94.32	93.99	93.77
	Kappa	95.26	95.74	95.54	**95.86**	95.63	94.12	95.53	95.66	94.74
NC16	OA	94.96	96.13	95.88	96.21	**96.53**	96.46	96.04	96.05	96.13
	AA	90.35	90.85	89.84	89.83	**90.28**	91.28	89.77	90.46	89.54
	Kappa	93.08	94.66	94.31	94.77	**95.21**	95.11	94.54	94.55	94.67
NC13	OA	75.96	76.73	76.67	76.58	**78.27**	76.83	77.14	76.97	76.36
	AA	82.78	81.58	81.59	82.35	**82.84**	82.93	83.44	84.19	82.94
	Kappa	71.47	72.39	72.33	72.23	**74.21**	72.50	72.88	72.69	71.98

TABLE VIII

COMPUTING TIME OF DIFFERENT METHODS ON THREE DATASETS

Data	Time	Methods							
		1DCNN	CDCNN	3DCNN	SSRN	DBMA	DBDA	HybridSN	Our method
NC12	Training(s)	**109.17**	596.06	1116.16	2093.90	2524.14	2406.79	1483.81	667.55
	Test(s)	**25.61**	725.18	870.14	1115.46	1398.42	1357.04	1024.44	715.53
NC16	Training(s)	**33.30**	188.25	340.01	671.18	777.65	750.81	459.14	206.15
	Test(s)	**14.93**	564.57	639.68	760.17	878.44	906.06	744.62	545.97
NC13	Training(s)	**32.76**	193.56	344.16	666.15	803.99	772.24	471.93	203.15
	Test(s)	**12.03**	459.25	581.67	645.91	794.90	792.30	639.35	502.38

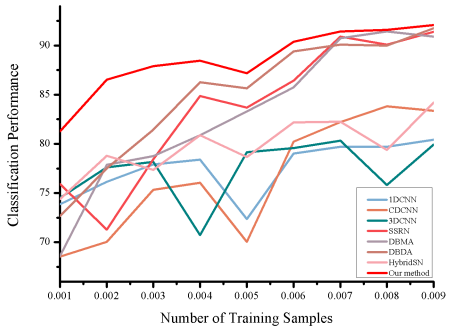


Fig. 12. Classification accuracies of different methods on the NC13 dataset with different numbers of training samples.

is the fastest method among all compared methods because it only processes the spectral information without considering spatial information. Therefore, its classification performance is significantly inferior to other methods. Compared to other spectral–spatial classification methods, our method takes less time on testing for the YRD datasets.

Analyzing the Influence of the Number of Samples: From Fig. 12, it is clear to observe that the classification performance of all methods improves as the number of training sets increases. In addition, our method always obtains the best classification performance when the number of training samples is varying.

V. CONCLUSION

In this work, we propose an MGSSAN for mapping wetland species, which mainly consists of two key steps. First, we develop two-channel CNNs with the residual structure to extract the discriminative features from the input images, and a global spectral–spatial mechanism is designed to preserve useful features and suppress useless ones. Furthermore, the ERW model is used to optimize the initial classification map by using the correlation between neighboring pixels. Experiments performed on the created database demonstrate that the proposed method can always obtain better classification performance than other state-of-the-art classification methods.

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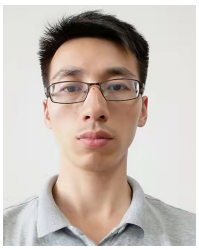
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